MiNA – Real Estate Wealth (MiNA-Rew)

Technical Appendix and Data Description

Micro-based National Accounts for the Swedish Household Sector

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Abstract

This appendix describes the construction of individual-level real estate wealth for Swedish households covering five property categories — single-family houses, holiday homes, multi-dwelling rental buildings, agricultural land, and industrial properties — over the period 1999–2018. The fundamental measurement problem is that directly observed micro-level market values, available from official administrative sources through 2007, ceased to exist when the Swedish wealth tax was abolished. We overcome this gap by estimating market-to-assessed-value multipliers from arm's-length property transactions recorded in the Real Estate Property Prices and Title Deeds Registry (*Fastighetsprisregistret*; SCB (2020a)) and applying them to the complete ownership stock recorded in the Real Estate Property Valuation Registry (*Fastighetstaxeringsregistret*; SCB (2020b)). Geographic heterogeneity in multipliers is captured through spatially constrained clustering of municipalities, following the max-p regions approach of Duque et al. (2012). The construction resolves three substantial data-quality challenges in the underlying registers before estimating multipliers: unstable identifiers, incomplete transaction parties, and misaligned ownership timing. We estimate market price totals per property owner per year using valuation multipliers that are generated over time-varying, multiplier-homogeneous spatial clusters.

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1 Introduction and Scope

Real estate is among the most widely held and economically significant assets on Swedish household balance sheets. Constructing a comprehensive household wealth panel therefore requires market-value estimates for non-apartment real estate holdings across the full population of owners and over multiple decades. This appendix documents the construction of the real estate wealth component $W_{i,t}^{\mathrm{Re}}$, the individual i's estimated market value of non-apartment real estate in year t.

The core measurement problem is a data discontinuity in 2007. Until that year, Statistics Sweden published individual-level market-value estimates as part of the annual wealth tax assessment process (*Förmögenhetsregistret (FORM)*; Statistics Sweden (2007)), providing a benchmark for the five property categories we study. After 2007, no official micro-level market-value series exists for these property categories. The target of this appendix is therefore the post-2007 market value of directly held non-apartment real estate. FORM enters in this context because it provides an observed benchmark during the overlap period through 2007, even though its exact replication is not the scientific objective.

We address a standard stock-flow measurement problem in administrative wealth data. The ownership register contains assessed values for the stock of real estate; the transaction register contains market prices for the transacted subset. We use the transaction data to estimate market-to-assessed-value multiplier averages over region-property category clusters and transport those multiplier estimates to the full stock of real estate units, recovering market values for all relevant holdings including after the official series ends. This approach follows the broad logic used by Statistics Sweden in constructing the historical market-value series — multiplying assessed values by purchase-price multipliers derived from transactions — and extends it with a more explicit, data-driven approach to spatial aggregation.

The project rests on two administrative registers. The ownership stock comes from the Real Estate Property Valuation Registry (*Fastighetstaxeringsregistret (FTR)*; Statistics Sweden (2020b)), which records every individual's registered share in each real estate tax unit, together with the official tax-assessed value and property type. Market price information comes from the Real Estate Property Prices and Title Deeds Registry (*Fastighetsprisregistret (FPR)*; Statistics Sweden (2020a)), which records purchase prices and purchase-price-to-assessed value coefficients for registered title transfers. Before market prices can be estimated from these sources, three substantial data-quality challenges must be resolved: a structural change in tax-unit identifiers between 2003 and 2004, incomplete transaction-party information in older records, and a mismatch between the ownership register's annual snapshots and the actual timing of property transfers.

Section 2 describes the data sources and the study sample. Section 3 defines the target

variable and key intermediate objects. Section 4 documents the construction procedure. Section 5 describes the final output and validation.

2 Data Sources and Sample Construction

2.1 Overview of Data Sources

The primary stock register is the Real Estate Property Valuation Registry (Statistics Sweden, 2020b), which spans the years 1998–2018. It provides the annual ownership composition of every tax unit in Sweden, recording each co-owner's personal or organizational identifier, registered ownership share, property type code, official tax-assessed value, and municipality. The registry is organized at two levels: an owner-by-tax-unit panel and a property-by-tax-unit panel; both are used in the construction.[1]

The primary transaction register is the Real Estate Property Prices and Title Deeds Registry (Statistics Sweden, 2020a), covering 1998–2018. It records the purchase price, the tax-assessed value at time of transfer, and the resulting purchase-price coefficient for registered title transfers, together with transaction party identifiers, date, property type, and municipality. FPR is delivered as a relational register, including transaction-level and party-level files, as well as property-level and tax-unit-level transfer files, which are used to interpret multi-property and multi-tax-unit transactions and to motivate the sample restrictions below. For the pre-2008 validation period, we draw on the annual Wealth Tax Registry (Statistics Sweden, 2007), which contains official market-value estimates and official tax-value estimates at the individual by property category level, 1999–2007. These files do not enter the production construction; they serve as an external benchmark for the multiplier-based estimates over the overlap period.

For geographic clustering, we use the published Swedish geographic datasets (Statistics Sweden, 2021b) together with a queen-contiguity spatial-weights matrix derived from those boundaries. Published municipality-level purchase-price coefficients (Statistics Sweden, 2025d,c,a,b) serve as a secondary reference for multiplier comparison. For aggregate validation after 2007, we use published National Accounts balance-sheet series for the household sector (Statistics Sweden, 2021a).

[1] A *tax unit* (*taxeringsenhet*) is the administrative property-assessment object, generally corresponding to a property or a group of properties with the same owner and use. A single tax unit can contain one or more underlying properties and, in more complex cases, one or more valuation units, but it is assessed jointly and carries the tax-assessed value to which the valuation multiplier is applied. The tax-assessed value (*taxeringsvärde*) is set to approximately 75% of the estimated market value at a reference date tied to the latest assessment cycle; assessment cycles differ by property type.

Table 1: Source data used in the construction

Source	Provider	Years	Information contributed
Real Estate Property Valuation Registry (FTR)	Statistics Sweden (2020b)	1998–2018	Ownership stock, assessed values, property types, shares
Real Estate Property Prices and Title Deeds Registry (FPR)	Statistics Sweden (2020a)	1998–2018	Transaction prices, purchase-price coefficients, parties
Wealth Tax Registry	Statistics Sweden (2007)	1999–2007	Official market- and tax-value estimates by property category
Municipality boundaries	Statistics Sweden (2021b)	Static	Municipal geometry and contiguity weights for clustering
Published multiplier tables	Statistics Sweden (2025d,c,a,b)	Varies	Official purchase-price coefficients by region and property category
National Accounts (ESA2010)	Statistics Sweden (2021a)	1980–2020	Aggregate household-sector real estate wealth

2.2 Sample Definition and Construction

The study population comprises all individuals and legal entities registered as owners of non-apartment real estate in Sweden in at least one year between 1999 and 2018. The five property categories covered are: single-family houses (*småhus*), holiday homes (*fritidshus*), multi-dwelling rental buildings (*hyreshus*), agricultural land (*lantbruk*), and industrial properties (*industri*). Cooperative apartments, which are documented in a companion appendix (Sodini and MiDa, 2026), are excluded.

The unit of observation is an individual owner of a given tax unit in a given year. All data sources described above were linked to the FTR to construct the final panel, using the harmonized tax-unit identifier described in Section 4.2. Holiday home transactions are available through 2014; properties in this class are valued after 2014 using single-family-house multipliers as a proxy.[2]

3 Key Definitions and Base Variables

This section fixes the notation used throughout the appendix to describe the estimation of market values of directly held non-apartment real estate after 2007.

Tax unit and property category. Let j index a tax unit and t a calendar year. A tax unit is the natural intermediate valuation object because it is the administrative unit for assessed values in the FTR and because FPR transaction coefficients are defined relative to assessed values for transferred tax units. Tax units are grouped into five broad property categories, denoted $c(j)$: single-family houses, holiday homes, multi-dwelling rental buildings, agricultural properties, and industrial properties.

Assessed value. Let $\tau^{\mathrm{Re}}_{j,t}$ denote the FTR-recorded tax-assessed value of tax unit j in year t. This assessed value is the base to which transaction-derived market-to-assessed-value multipliers are applied. It is not itself a contemporaneous market value.

[2]SCB discontinued separate holiday-home transaction reporting in the property price register after 2014.

Ownership share. Let $s^{\mathrm{Re}}_{i,j,t}$ denote owner i's ownership share in tax unit j in year t. The raw ownership records are repaired before final valuation because transaction-party information and annual ownership timing are incomplete in parts of the data. Section 4.2 describes those repairs.

Transaction coefficient and valuation multiplier. Let $\kappa^{\mathrm{Re}}_{\ell}$ denote the purchase-price coefficient for admissible transaction ℓ, defined as the ratio of the observed transaction price to the relevant assessed-value denominator. We aggregate these transaction coefficients into market-to-assessed-value multiplier estimates $\widehat{\mu}_{t,c,r}$, where r denotes the valuation region used for year t and property category c. The transaction sample and the regional multiplier construction are described in Sections 4.3 and 4.4.

Tax-unit valuation identity. For any chosen multiplier specification, the imputed market value of tax unit j in year t is

$$\widehat{x}^{\mathrm{Re}}_{j,t} = \tau^{\mathrm{Re}}_{j,t}\, \widehat{\mu}_{t,c(j),r(j,t)}, \tag{1}$$

where $r(j,t)$ is the valuation region assigned to tax unit j. Equation (1) is the core valuation identity. The remaining sections explain how the ownership panel, transaction sample, regional multipliers, and validation objects are constructed.

4 Real Estate Valuation Procedure

4.1 Overview of the Procedure

We proceed in three construction steps and one benchmarking step. First, we harmonize the ownership-stock and transaction registers into a coherent owner–tax-unit–year panel. Second, we estimate market-to-assessed-value multipliers from arm's-length transactions, with separate treatment of property category, year, and geography. Third, we apply those multipliers to the full FTR ownership stock to produce tax-unit-level estimated market values, which are then allocated to owners in Section 5. We then benchmark the resulting series against overlap-period FORM values and post-2007 National Accounts aggregates.

4.2 Constructing a Coherent Owner–Tax-Unit Panel

Before valuation, we repair three data problems that would otherwise distort either the ownership stock or the link between the stock and transaction records. First, the tax-unit identifier changed between the 2003 and 2004 assessment cycles. We therefore construct a harmonized identifier that bridges the pre-2004 composite key to the post-2004 tax-unit identifier inside both the FTR and FPR.

Second, before May 2004 the FPR includes only one individual per side of each transaction, even when multiple co-owners were involved. The project recovers missing buyers, sellers, and ownership shares by combining the observed FPR parties with FTR co-ownership patterns and nearby FPR events on the same tax unit. The recovery distinguishes ordinary transfers from transfers within an existing co-owner set, where a person may legitimately appear on both sides of a transaction. Recovered identities and shares are retained with imputation flags.

Third, FTR is an annual stock register, whereas FPR records transaction timing. We use FPR acquisition and sale information to align annual ownership spells more closely with observed transfer timing. This step reconstructs the person–tax-unit–year exposure panel to which the valuation in equation (1) is later attached.

4.3 Transaction Sample for Multiplier Estimation

The multiplier $\widehat{\mu}^{v}_{t,c,r}$ must be estimated from a subset of transactions whose prices and assessed values can be interpreted as measuring the same economic object. Following SCB property-price-statistics material and the FPR metadata underlying the published transaction statistics (Statistics Sweden, 2020a, 2025d,c,a,b), we restrict FPR to arm's-length market transfers: ordinary purchases, transfers covering the whole assessed tax unit / whole transferred object, comparable assessed values, and standard freehold transfer form. Transactions with extreme purchase-price coefficients are excluded.[3]

These filters enforce the requirement that the purchase price and the assessed value should refer to the same object, so that a transaction-specific multiplier can be meaningfully transported to the ownership stock. This requirement is especially important for multi-tax-unit transfers: in some FPR transaction-level records, the reported tax attributes refer to a selected main tax unit rather than to the full set of tax units included in the transfer. Such records can make the price, assessed value, type code, and transfer scope inconsistent and require imposing whole-object and comparability restrictions.

Filtering alone does not address all problems around mismatches in denominators (assessed values) between the FPR and FTR. For a limited set of early-year property category combinations, the coefficient denominator in the FPR can differ from the FTR assessed-value concept to which the multiplier is later applied due to base-value or recalculation differences. In those cases, we recompute the transaction coefficient using the FTR-side assessed-value denominator before averaging multipliers.

[3] Observations with purchase-price coefficient below 0.8 or above 6 are excluded for single-family houses, holiday homes, multi-dwelling buildings, and industrial properties; the lower bound is 0.5 for agricultural land.

4.4 Multiplier Estimation and Spatial Clustering

The multiplier varies along three dimensions: time, because transaction prices move year to year while assessed values are fixed between assessments; property category, because different property types have different price dynamics; and space, because real-estate markets are spatially segmented. We treat year and property category as fixed organizing dimensions and estimate six alternative constructions of the geographic dimension.

For a set $\mathcal{C}_{t,c,r}$ of admissible transactions in year t, property category c, and region r, the regional multiplier is the geometric mean of the transaction-level purchase-price coefficients:

$$\widehat{\mu}^{v}_{t,c,r} = \exp\left(\frac{1}{N_{t,c,r}} \sum_{\ell \in \mathcal{C}_{t,c,r}} \log \kappa^{\mathrm{Re}}_{\ell}\right), \qquad N_{t,c,r} = |\mathcal{C}_{t,c,r}|. \tag{2}$$

Here $\kappa^{\mathrm{Re}}_{\ell}$ denotes the transaction coefficient after the sample restrictions and denominator corrections described in Section 4.3. Using the geometric mean reduces sensitivity to extreme transactions that remain after filtering.

The choice of geographic aggregation level involves a standard bias–variance trade-off: municipality-level estimation captures spatial heterogeneity faithfully but yields noisy estimates in sparsely traded areas, while national-level pooling is precise but ignores genuine regional variation. The first two variants are therefore transparent benchmarks: a fully pooled national-level estimate and a fully disaggregated municipality-level estimate. The remaining four use the max-p regionalization algorithm of Duque et al. (2012) to group municipalities into spatially contiguous clusters.

Let $\mathcal{R}_{t,c}$ denote a partition of municipalities into contiguous regions for year t and property category c. The max-p step searches for a partition that maximizes the number of feasible regions subject to a minimum transaction-count requirement:

$$\max_{\mathcal{R}_{t,c}} |\mathcal{R}_{t,c}| \quad \text{subject to} \quad N_{t,c,r} \geq \underline{N} \text{ and } r \text{ is contiguous for every } r \in \mathcal{R}_{t,c}. \tag{3}$$

We set $\underline{N} = 25$. Among feasible regionalizations, the algorithm favors within-region homogeneity in a chosen municipality-level characteristic $z_{k,t,c}$:

$$H(\mathcal{R}_{t,c}; z) = \sum_{r \in \mathcal{R}_{t,c}} \sum_{k \in r} \omega_{k,t,c} \left\| z_{k,t,c} - \bar{z}_{r,t,c} \right\|^2, \tag{4}$$

where k indexes municipalities, $\omega_{k,t,c}$ is a transaction-count weight, and $\bar{z}_{r,t,c}$ is the regional weighted mean. In the multiplier-homogeneity variants, $z_{k,t,c}$ is the municipality mean purchase-price coefficient. In the centroid variants, $z_{k,t,c}$ is based on municipality centroid coordinates, producing more geographically compact regions. Multiplier-based clustering is more directly tied to valuation error; centroid-based clustering is a geographic alternative.

The four max-p variants cross two temporal structures with these two homogeneity

criteria. In the updated variants, each municipality can receive a time-varying assignment $r_{t,c}(k)$, and the natural size requirement is

$$N_{t,c,r} \geq \underline{N} \qquad \text{for all } t, c, r. \tag{5}$$

In the fixed variants, each municipality receives a time-invariant assignment $r_c(k)$. A fixed partition still needs to meet the annual minimum-size constraint. We therefore build the size criterion for the fixed-region variants from annual transaction counts of years with few transactions rather than from cumulative counts, so that a cluster is not accepted simply because it has many transactions in some years and too few in others. Yearly updated regions avoid that fixed-partition constraint but require every municipality to remain in the annual clustering problem, including municipality-year cells with no admissible transactions. Clusters are defined separately for each of the five property categories.

Table 2: Multiplier estimation variants

Variant	**Method**	**Spatial structure**	**Time structure**
1	National pool	None (national)	Annual
2	Municipality	Municipality	Annual
3	Max-p (multiplier)	Contiguous clusters	Time-constant
4	Max-p (centroid)	Contiguous clusters	Time-constant
5	Max-p (multiplier)	Contiguous clusters	Time-dynamic
6	Max-p (centroid)	Contiguous clusters	Time-dynamic

Missing multiplier values — for example, municipalities not represented in the transaction data for a given property category — receive a municipality-level all-time mean as a final fallback. For holiday homes after 2014, when separate holiday-home transaction reporting is no longer available, the construction uses single-family-house multipliers as a proxy.

4.5 Benchmarking and Validation

4.5.1 Overlap-Period Validation Against FORM

The temporal overlap with FORM provides a micro-level benchmark during 1999–2007. We compare predicted market values from valued FTR to observed official values at the individual-by-property-category-by-year level. These differences are between two deterministic administrative valuation procedures rather than classical forecast errors against noisy truth.

To construct the validation object, we infer a crosswalk from FTR type codes to FORM categories by linking FORM and FTR, restricting to observations with wealth in only

one FORM category, and identifying the dominant type-code for each FORM category. Let $g(\cdot)$ denote the resulting mapping from FTR type code to FORM category f. For individual i, year t, and FORM category f, define the relevant valued-FTR holdings as

$$\mathcal{I}_{i,t,f} = \left\{ u : \begin{array}{l} \text{owner}(u) = i, \\ \text{year}(u) = t, \\ g\Big(\text{type}_u\Big) = f \end{array} \right\}. \tag{6}$$

We aggregate tax-unit predictions to the FORM property category using

$$\widehat{x}^{v}_{i,t,f} = \sum_{u \in \mathcal{I}_{i,t,f}} \widehat{x}^{v}_{u}. \tag{7}$$

Let $x_{i,t,f}$ denote the observed FORM value. We evaluate each variant using both absolute and relative validation errors:

$$e^{\text{abs},v}_{i,t,f} = \widehat{x}^{v}_{i,t,f} - x_{i,t,f}, \tag{8}$$

$$e^{\text{rel},v}_{i,t,f} = \frac{\widehat{x}^{v}_{i,t,f} - x_{i,t,f}}{x_{i,t,f}} \cdot 100\%. \tag{9}$$

4.5.2 Aggregate Validation Against National Accounts

For the post-2007 period, aggregate predicted values are compared against published National Accounts balance-sheet series for the household sector (Statistics Sweden, 2021a), providing a macro-level external benchmark precisely where FORM is no longer available. At the property category level, the aggregate predicted value can be written as

$$\widehat{X}^{v}_{t,c} = \sum_{j \in \mathcal{S}_{t,c}} \widehat{x}^{\text{Re},v}_{j,t} = \sum_{r} \widehat{\mu}^{v}_{t,c,r} \sum_{j \in \mathcal{S}_{t,c,r}} \tau^{\text{Re}}_{j,t}, \tag{10}$$

where $\mathcal{S}_{t,c}$ is the set of tax units in year t and property category c, and $\mathcal{S}_{t,c,r}$ is the corresponding subset in region r. This decomposition makes clear that aggregate movements reflect both the multiplier component and the assessed-value component. The National Accounts comparison is useful but not exact, especially for industrial/commercial property, where the FTR/FPR property category concept and the household-sector National Accounts concept do not line up perfectly.

5 Final Construction of the Target Variable

The output dataset contains one record per individual owner per tax unit per year, covering 1999–2018. For each of the six multiplier variants $v \in \{1, \ldots, 6\}$, we compute a tax-unit-level estimated market value by applying equation (1) with the multiplier definition corresponding to variant v. We then allocate that value to owners using ownership

shares:

$$W_{i,t}^{\mathrm{Re},v} = \sum_{j \in \mathcal{P}_{i,t}^{\mathrm{Re}}} s_{i,j,t}^{\mathrm{Re}} \, \widehat{x}_{j,t}^{\mathrm{Re},v}, \tag{11}$$

where $\mathcal{P}_{i,t}^{\mathrm{Re}}$ is the set of tax units in which owner i holds a registered share in year t. The panel covers all registered owners of non-apartment real estate in Sweden and includes audit flags indicating whether each ownership record, assessed value, ownership share, or multiplier was imputed and by which route.

The preferred specification is Variant 5 (time-dynamic, multiplier-homogeneity clustering) as it delivers the lowest bias and mean absolute error after trimming the validation-error distribution in the overlap-period comparison. The remaining five variants are preserved in the output to support sensitivity analysis.

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A Glossary

Term	Definition
Fastighetstaxeringsregistret (FTR)	Real Estate Property Valuation Registry. Annual administrative register maintained by Statistics Sweden containing the ownership composition, tax-assessed values, and property attributes of real estate tax units in Sweden.
Fastighetsprisregistret (FPR)	Real Estate Property Prices and Title Deeds Registry. Administrative register of registered property title transfers covered by FPR, containing transaction prices, tax-assessed values, purchase-price coefficients, and party identifiers.
Taxeringsvärde (tax-assessed value)	The official assessed value of a property determined by Statistics Sweden in its periodic assessment cycle, intended to represent approximately 75% of market value at a reference date. Used as the tax base and as the denominator in computing purchase-price coefficients.
Basvärde (base value)	An alternative assessed value used by SCB as the multiplier denominator for certain property categories and years in the published purchase-price coefficients; distinct from *taxeringsvärde* for some asset-class and year combinations.

Term	Definition
FORM	Annual individual-level market-value estimates for real estate holdings, published by Statistics Sweden as part of the wealth tax assessment process, 1999–2007.
Taxeringsenhet (tax unit)	The administrative unit of property taxation in Sweden. A single physical property may correspond to one or more tax units; each tax unit carries its own assessed value and ownership register. For multiplier estimation, a tax unit enters the broad property class implied by its FTR type code.
Typkod	Property type code used in FTR to classify tax units by land use and structure type; the primary variable used to link tax units to the broad classes used in multiplier estimation and validation.
Fritidshus	Holiday home (vacation property).
Hyreshus	Multi-dwelling rental apartment building.
Industri	Industrial or commercial real estate.
Lantbruk	Agricultural land and associated buildings.
Småhus	Single-family house (one- or two-dwelling building).
SCB (*Statistiska centralbyrån*)	Statistics Sweden, the national statistical agency and custodian of the administrative registers used in this construction.
Max-p regions	A spatially constrained regionalization algorithm that maximizes the number of contiguous regions while requiring each region to meet a minimum size constraint (Duque et al., 2012).